



# A kinetic model to study the regulation of $\beta$ -catenin, APC, and Axin in the human colonic crypt

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**Abstract** The Wnt/ $\beta$ -catenin pathway plays a crucial role in stem cell renewal and differentiation in the normal human colonic crypt. The balance between  $\beta$ -catenin and APC along the crypt axis determines its normal functionality. The mechanism that deregulates this balance may give insight into the initiation of colorectal cancer. This is significant because the spatial dysregulation of  $\beta$ -catenin by the mutated tumor suppressor gene/protein APC in human colonic crypts is responsible for the initiation and growth of colorectal cancer. We consider a regulatory function that promotes APC synthesis within the cell and its effect on the accumulation of the Wnt target protein,  $\beta$ -catenin. It is evident that an APC gradient exists along the crypt axis; however, the mechanism by which APC expression is regulated within the cell is not well known. We investigate the dynamics of an APC regulatory mechanism with an increased level of Axin at the subcellular level. Model output shows an increase of APC for a diminished Wnt signal, which explains the APC gradient along the crypt. We find that the dynamic interplay between  $\beta$ -catenin, APC, and Axin produces oscillatory behavior, which is controlled by the Wnt stimulus. In the presence of reduced functional APC, the

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oscillations are amplified, which suggests that the cell remains in a more proliferative state for longer periods of time. Increased Axin levels (typical of mammalian cells) reduce oscillatory behavior and minimize the levels of  $\beta$ -catenin within the cell while raising the levels of APC.

**Keywords** Colonic crypt · APC regulation · Wnt pathway · Colorectal cancer · Multiple timescale analysis

**Mathematics Subject Classification** 92B05 · 92C37 · 92C40 · 92C42

## 1 Introduction

Cellular signaling pathways involve a cascade of biochemical reactions initiated by an extracellular stimulus that induce a physiological response important in embryonic development and cell function. Often, abnormal regulation of the pathway components lead to developmental defects as well as cancerous states (Weinberg 2007; Nusse 2005; Clevers and Nusse 2012). Therefore, it is important to understand the underlying mechanisms responsible for dysfunctional pathways. The goal of our study is to understand the dynamics of Wnt signaling and how it changes in response to reduced APC levels.

The Wnt protein ligand is one such stimulus that initiates the Wnt signaling pathway by binding to a Frizzled (Frz) receptor on the cell membrane. The canonical Wnt signaling pathway plays an important role in regulating the amount of  $\beta$ -catenin, a key intracellular component of Wnt signaling, whose activity promotes cellular proliferation and cell-to-cell adhesion (Clevers and Nusse 2012; Lloyd-Lewis et al. 2013; Kikuchi and Yamamoto 2007). In the absence of Wnt, a destruction complex made up of adenomatous polyposis coli (APC), Axin, and the enzyme GSK-3 $\beta$  phosphorylates  $\beta$ -catenin, which quickly leads to proteolysis (Weinberg 2007; Aberle et al. 1997; Latres et al. 1999; Liu et al. 1999). In the presence of Wnt, the destruction complex is deactivated by the intermediate protein Dishevelled (Dsh) and accumulation of  $\beta$ -catenin occurs within the cytoplasm. Upon translocating to the nucleus,  $\beta$ -catenin binds with transcription factors (TCF) which drives the expression of a variety of genes, particularly those responsible for cell proliferation (Weinberg 2007; Vries et al. 2010). Wnt induced  $\beta$ -catenin activity is directly regulated by the APC protein since APC is part of the destruction complex. However, in the upper crypt, at a position distant from the Wnt stimulus at the crypt bottom, APC concentrations are higher (Boman and Fields 2013). Furthermore, APC plays a role in spindle orientation and microtubule function during mitosis (Boman and Fields 2013). Therefore, APC is an active factor in cell differentiation.

Several mathematical models have been created to describe the intracellular dynamics that make up the Wnt pathway. Lee et al. produced a quantitative model in Lee et al. (2003) that includes the core proteins of the Wnt pathway. A system of non-linear ODEs in APC, Axin, GSK-3 $\beta$ ,  $\beta$ -catenin, and TCF concentrations describe the accumulation of nuclear  $\beta$ -catenin via an extracellular Wnt stimulus. Lee and coworkers derive quantitative data from the *Xenopus* egg extract to define specific protein parameters of the system; however, a recent study by Tan et al. suggests that

these measurements may be different compared to protein levels in mammalian cell cultures (Tan et al. 2012). Cho et al. (2006) extend this model to include an induced synthesis of Axin. Cho et al. also investigated changes in the system due to APC gene mutations. Wawra et al. (2007) expanded on Cho's model to include feedback mechanisms and time delays in the synthesis of Axin via the intermediate product Axin 2 RNA. They also investigate the interference of the DKK1 protein with Dsh using a time delay. They were able to show oscillations in the model can occur under particular circumstances. Furthermore, Wawra et al. perform an extended analyses of the model to show its robustness under parameter perturbation. Finally, Mirams et al. perform a multiple timescale analysis of the Lee–Heinrich model to consider the long term accumulation of  $\beta$ -catenin (Mirams et al. 2010). In this study, the original system is effectively reduced to one equation. This is especially useful in multi-scale modeling where a simpler model of subcellular dynamics can be incorporated in a larger model to describe cell properties, as in Leeuwen et al. (2010), Murray et al. (2009), Murray et al. (2010) and Murray et al. (2011).

Understanding the regulation of  $\beta$ -catenin and APC via Wnt is especially important in the colon. The epithelium of the large intestine contains approximately  $10^7$  finger-like invaginations known as colonic crypts. The function of the crypt is to regulate the development of healthy cells and to protect stem cells from the toxic materials in the colon (Weinberg 2007). Approximately 3000 cells make up the human colonic crypt, with about 20 stem cells residing at the base (Potten and Loeffler 1990; Nishimura et al. 2003; Huang et al. 2009; Boman and Huang 2008). Stem cells divide, on average, asymmetrically, thereby generating a stem cell population and transient amplifying cell population. Transient amplifying cells migrate up the crypt while dividing and are eventually extruded into the intestinal lumen. In essence, the progeny of the stem cells make up the majority of the epithelial lining of the colon, which is renewed every 3–5 days. Wnt ligands are produced near the crypt base by stromal cells in the extracellular matrix (Weinberg 2007). This contributes to a gradient of Wnt factors along the major axis of the crypt. Furthermore, inverse gradients of intracellular  $\beta$ -catenin and APC are observed (Boman and Fields 2013). In experimental results conducted by Boman et al. and Kosinski et al., the amount of intracellular APC increases nearly fivefold as measured from the bottom of the crypt to the top (Zhang et al. 2010; Kosinski et al. 2007). Although a gradient in APC is observed, little is known about the molecular regulatory mechanism that generates this gradient; however, cellular proliferation appears to drive production of APC (Boman and Fields 2013). It is thought that the inverse gradients create a zone of rapid proliferation approximately one-third of the way up the crypt that drives most of cell movement and crypt renewal (Boman and Fields 2013). The inverse profiles of extracellular Wnt factors and intracellular APC proteins along the major crypt axis are responsible for position-dependent rates of cell proliferation, differentiation, and apoptosis. Therefore, understanding the pathway dynamics is essential to understanding mechanisms that maintain healthy colonic crypt organization as well as those that underlie colorectal cancer initiation.

About 90% of all sporadic colon carcinomas are initially characterized by an accumulation of  $\beta$ -catenin even in the absence of Wnt signaling (Weinberg 2007). This imbalance is due to the systematic inactivation of the APC gene, resulting in a loss of function of the APC protein. It is well known that the APC protein loses its func-

tion through a two-hit mutational event that often leads to expression of a truncated APC protein lacking specific binding sites (Lloyd-Lewis et al. 2013; Boman and Fields 2013; Cho et al. 2006; Wodarz and Komarova 2005; Fearnhead et al. 2001; Gaspar and Fodde 2004; Albuquerque et al. 2002). This mimics the effect of Wnt pathway activation that extends further up the crypt thereby creating an accumulation of  $\beta$ -catenin within the epithelial cells. In effect, the delicate balance between cell proliferation, differentiation, and apoptosis, is disrupted creating an abundance of progenitor-like phenotypes, which can be observed in most hereditary cases of familial adenomatous polyposis (FAP) (Weinberg 2007; Boman et al. 2007). In general, the transition between cell phenotypes is delayed, which shifts the population of stem-like cells toward the crypt top, eventually leading to their overpopulation. Adenomatous precancerous polyps appear in the intestinal epithelium and progression of these tumors is accelerated by crypt fission (Boman and Fields 2013; Wasan et al. 1998).

In this study, we investigate the effects of APC regulation at the subcellular level and how this regulation may be the source of counter-current-type mechanisms responsible for normal crypt homeostasis. In Sect. 2, we create an intracellular model based on an expansion of the Lee–Heinrich ODE model that incorporates a  $\beta$ -catenin and TCF dependent synthesis of free APC. In the same section, we consider an increased level of Axin within the system, and carry out a multiple timescale analysis to determine the effects of Axin on APC and  $\beta$ -catenin on the long timescale. In Sect. 3, we present a sub-case of the model that isolates the effects of the APC regulatory mechanism. Further, we consider the reduced leading-order system that describes the dynamics of  $\beta$ -catenin factors and APC that could potentially be used for multi-scale models in the crypt. Finally, in Sect. 4 we discuss the implications of this new model and its influence on the Wnt pathway in the human colon.

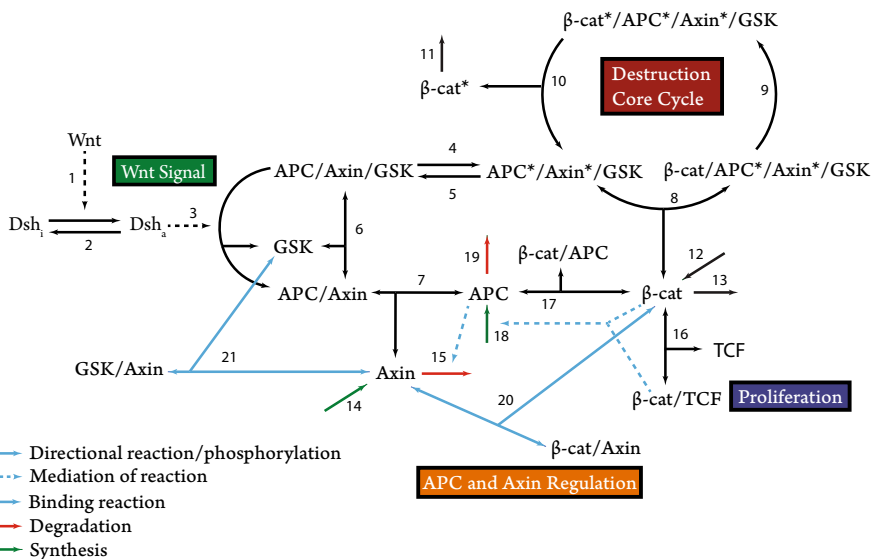
## 2 Regulation of APC and Axin

The ODE system presented by Lee et al. (2003) was derived from biological experiments on the *Xenopus* egg extract. This model assumes the cytoplasm of the cell is well mixed, so a system of 15 time-dependent ordinary differential equations is formulated. The model includes the following components that are essential to the Wnt cascade: Wnt, Dsh, GSK-3 $\beta$ , APC, Axin,  $\beta$ -catenin, and TCF. The model also assumes the conservation of APC, TCF, GSK-3 $\beta$ , and Dsh. Although the amount of APC is relatively constant during Wnt stimulation, it is known that APC varies along the human crypt axis. However, it is not well known how APC is regulated, but differentiated cells further up the crypt tend to have more intracellular APC (Boman and Fields 2013). Downstream Wnt factors such as  $\beta$ -catenin and  $\beta$ -catenin/TCF promote cell proliferation, but when cells proliferate, their progeny gain APC. Therefore,  $\beta$ -catenin down-regulates itself through the synthesis of APC due to cell proliferation. We construct a model extension to the ODE system of Lee et al. to incorporate this feedback.

The regulation of both APC and Axin play an important role in the Wnt signaling cascade. A feedback loop involving the degradation of Axin by APC is thought to limit the accumulation of Axin in the presence of APC mutations (Lee et al. 2003). In this

sense, when APC is lacking, Axin is recruited to help form the destruction complex and degrade  $\beta$ -catenin. Furthermore, in mammalian intestinal cell lines, the amount of Axin has been measured and found to be higher in concentration to that found in the *Xenopus* egg extract (Tan et al. 2012). As biological research progress yields further insight into the Wnt signaling cascade, further adjustments and assumptions can be made to provide a more accurate mathematical model.

We consider other extensions to Lee's model that will provide a framework for cell cultures in mammals. This means, we do not assume Axin is small compared to other protein concentrations, and because of this, we include several additional binding reactions that involve Axin. Furthermore, we incorporate APC regulation with a feedback on Axin by APC, which will help to stabilize the amount of  $\beta$ -catenin in the cell. Figure 1 demonstrates the original reaction kinetics, denoted by black arrows, between each protein, protein complex, and phosphorylated protein/protein complex. The additional kinetics (blue, green, and red arrows) indicate our model extensions that reflect the mediated production of APC by  $\beta$ -catenin, the mediated destruction of Axin by APC, and the additional binding reactions with Axin. In the remainder of this section, we present a formulation, nondimensionalization, and multiple timescale analysis of the model with our additional reactions and end the section with a discussion.



**Fig. 1** The colored arrows represent added reactions to the Lee model. Reaction step 18 is a synthesis of APC (green arrow) mediated by  $\beta$ -catenin and  $\beta$ -catenin/TCF, depicted by the dashed-blue arrows. Reaction step 19 is a constant degradation of APC, denoted by the red arrow. These reactions are implemented to capture the mediated synthesis of APC by free and nuclear  $\beta$ -catenin. Binding reaction steps 20 and 21, depicted by the blue solid arrows, are additional reactions incorporated to capture the binding of Axin to GSK-3 $\beta$  and  $\beta$ -catenin if Axin is abundant in the cell. Also, we include an APC mediated degradation of Axin (reaction 15) as proposed by Lee et al. Black arrows represent all original reactions from Lee et al. Modified from Lee et al. (2003) (color figure online)

## 2.1 Model formulation

We consider the original set of 15 variables (Lee et al. 2003) that include the concentrations of proteins, protein complexes, phosphorylated proteins, enzymes, and transcription factors depicted in the reaction scheme (Fig. 1). We are assuming Axin may be more available within the cell since mammalian cell cultures express a larger concentration than that of the *Xenopus* egg extract (Tan et al. 2012). Under these conditions, the biology suggests that we include two more protein complexes,  $\beta$ -catenin/Axin and GSK-3 $\beta$ /Axin, which we denote by  $X_{16}$  and  $X_{17}$ , respectively (Lee et al. 2001). This gives a system of 17 equations in the variables  $X_i$  for  $i = 1, 2, \dots, 17$ . We employ the following set of equations:

$$\frac{dX_1}{dt} = -r_1 + r_2 \quad (1)$$

$$\frac{dX_2}{dt} = r_1 - r_2 \quad (2)$$

$$\frac{dX_3}{dt} = r_4 - r_5 - r_8 + r_{10} \quad (3)$$

$$\frac{dX_4}{dt} = -r_3 - r_4 + r_5 + r_6 \quad (4)$$

$$\frac{dX_5}{dt} = r_3 - r_6 \quad (5)$$

$$\frac{dX_6}{dt} = r_3 - r_6 + r_7 \quad (6)$$

$$\frac{dX_7}{dt} = -r_7 - r_{17} + r_{18} - r_{19} \quad (7)$$

$$\frac{dX_8}{dt} = r_8 - r_9 \quad (8)$$

$$\frac{dX_9}{dt} = r_9 - r_{10} \quad (9)$$

$$\frac{dX_{10}}{dt} = r_{10} - r_{11} \quad (10)$$

$$\frac{dX_{11}}{dt} = -r_8 + r_{12} - r_{13} - r_{16} - r_{17} \quad (11)$$

$$\frac{dX_{12}}{dt} = -r_7 + r_{14} - r_{15} \quad (12)$$

$$\frac{dX_{13}}{dt} = -r_{16} \quad (13)$$

$$\frac{dX_{14}}{dt} = r_{16} \quad (14)$$

$$\frac{dX_{15}}{dt} = r_{17} \quad (15)$$

$$\frac{dX_{16}}{dt} = \mathbf{r_{20}} \quad (16)$$

$$\frac{dX_{17}}{dt} = \mathbf{r_{21}}, \quad (17)$$

where  $r_j$  denotes reaction  $j$  in Fig. 1. New reactions are bolded in the equations above. Reactions 12 and 14 describe a source of protein or protein synthesis, which we assume leads to rate constants, denoted by  $v_{12}$  and  $v_{14}$ . Un-mediated protein degradations (reactions 11 and 13) are described as first order processes with rates  $r_i = k_i X_j$ . Phosphorylations by kinase (reactions 4 and 9), dephosphorylation by phosphatase (reaction 5), and the irreversible release of phosphorylated  $\beta$ -catenin (reaction 10) are also described by linear rates  $r_i = k_i X_j$ . Activation of Dsh by the Wnt signal (reaction 1) is described by  $r_1 = k_1 X_1 W$ , where  $W$  denotes the normalized concentration of the Wnt-ligand, i.e., a nondimensional forcing term. Inactivation of Dsh (reaction 2) is described by a first order process  $r_2 = k_2 X_2$ . Reaction 3, which is the Dsh mediated release of GSK-3 $\beta$ , is given by the rate equation  $r_3 = k_3 X_2 X_4$ , where  $X_2$  is not being consumed in the reaction, rather it acts as an activator. Finally, the binding processes of the scheme (reactions 6, 7, 8, 16, and 17) are described by the rate equations  $r_i = k_{+i} X_j X_\ell - k_{-i} \{X_j, X_\ell\}$ , where  $X_j$  and  $X_\ell$  denote the free concentrations of the binding partners and  $\{X_j, X_\ell\}$  is the product of the binding process between the two concentrations. To summarize, the following reaction rates,  $r_i$ , for  $i = 1, 2, \dots, 14, 16$ , and 17, are given below, where each  $k_i$  or  $v_i$  is a positive constant:

$$r_1 = k_1 X_1 W \quad (18)$$

$$r_2 = k_2 X_2 \quad (19)$$

$$r_3 = k_3 X_2 X_4 \quad (20)$$

$$r_4 = k_4 X_4 \quad (21)$$

$$r_5 = k_5 X_3 \quad (22)$$

$$r_6 = k_6 X_5 X_6 - k_{-6} X_4 \quad (23)$$

$$r_7 = k_7 X_7 X_{12} - k_{-7} X_6 \quad (24)$$

$$r_8 = k_8 X_{11} X_3 - k_{-8} X_8 \quad (25)$$

$$r_9 = k_9 X_8 \quad (26)$$

$$r_{10} = k_{10} X_9 \quad (27)$$

$$r_{11} = k_{11}X_{10} \quad (28)$$

$$r_{12} = v_{12} \quad (29)$$

$$r_{13} = k_{13}X_{11} \quad (30)$$

$$r_{14} = v_{14} \quad (31)$$

$$r_{16} = k_{16}X_{11}X_{13} - k_{-16}X_{14} \quad (32)$$

$$r_{17} = k_{17}X_7X_{11} - k_{-17}X_{15}. \quad (33)$$

We employ an important Wnt pathway component considered by [Lee et al. \(2003\)](#)—the feedback loop on the degradation of Axin by APC given by

$$r_{15} = \frac{k_{15}X_7X_{12}}{K_m + X_7}, \quad (34)$$

for positive constants  $k_{15}$  and  $K_m$ . Here,  $K_m$  is the maximum value that APC achieves within the crypt, namely  $K_m \approx 98$ , and  $k_{15} = 0.33$ , which is exactly double of the Lee–Heinrich model. In this way, the reaction rate  $r_{15}$  is the same as the original model when APC is maximized, but is small in the presence of APC mutations.

In the formulation of the model, Lee et al. assume that Axin and  $\beta$ -catenin are the only components produced and degraded. Furthermore,  $\beta$ -catenin is destroyed in two forms: after it is phosphorylated by the destruction complex and in its free state. Axin is only degraded in its free form. In our model, we include the assumption that APC can be produced within the cell. Because APC is no longer conserved, we cannot eliminate the variable  $X_7$  (APC) using a conservation equation. Furthermore, this assumption implies that the amount of APC varies substantially more than in the Lee–Heinrich model. This again is justified by the observable increase in the amount of APC in crypt cells anatomically away from the Wnt signal ([Boman and Fields 2013](#)). Reactions 18 and 19 represent the synthesis and degradation of APC, respectively, which are given by the functions

$$r_{18} = \frac{v_{18}}{1 + \frac{X_{14}}{K_t} + \frac{X_{11}}{K_b}}, \quad r_{19} = k_{19}X_7, \quad (35)$$

where  $v_{18}$ ,  $k_{19}$ ,  $K_t$ , and  $K_b$ , are positive, constant parameters. Here, we assume the synthesis is more rapid for less  $\beta$ -catenin, which explains the Hill-type expression in  $X_{14}^{-1}$  and  $X_{11}^{-1}$ . Furthermore, the total amount of free APC is degraded by a first order reaction  $k_{19}X_{17}$ . This is due to APC binding to other proteins within the cell such as EB1, which is essential to cytokinesis ([Boman and Fields 2013](#)).

Finally, we consider the two new equations, binding reactions 20 and 21, given as

$$r_{20} = k_{20}X_{11}X_{12} - k_{-20}X_{16} \quad (36)$$

$$r_{21} = k_{21}X_5X_{12} - k_{-21}X_{17}, \quad (37)$$

for positive constants  $k_{\pm 20}$  and  $k_{\pm 21}$ . These reactions describe the binding relationships between Axin and  $\beta$ -catenin and Axin and GSK-3 $\beta$  due to an abundance of Axin in the system.

We can eliminate a number of equations by identifying conserved quantities and making rapid equilibrium approximations. The concentrations of Dsh, GSK-3 $\beta$ , and TCF are conserved in the system and Lee et al. (2003) provide values for the total amounts of each conserved quantity denoted by  $Dsh^0$ ,  $GSK^0$ , and  $TCF^0$ , respectively. This gives the following three conservation equations:

$$X_1 + X_2 = Dsh^0 \quad (38)$$

$$X_3 + X_4 + X_5 + X_8 + X_9 + X_{17} = GSK^0 \quad (39)$$

$$X_{13} + X_{14} = TCF^0. \quad (40)$$

Other simplifications are implemented through rapid equilibrium assumptions. Lee and coworkers use the law of mass action to approximate the binding reactions 7, 8, 16, and 17, in which they provide estimates for the dissociation constants, denoted by  $K_7$ ,  $K_8$ ,  $K_{16}$ , and  $K_{17}$ , respectively. This yields the following four rapid equilibria equations:

$$\frac{X_7 X_{12}}{X_6} = K_7 \quad (41)$$

$$\frac{X_3 X_{11}}{X_8} = K_8 \quad (42)$$

$$\frac{X_{11} X_{13}}{X_{14}} = K_{16} \quad (43)$$

$$\frac{X_7 X_{11}}{X_{15}} = K_{17}. \quad (44)$$

Under the assumption that Axin may not be small, we include the binding reactions 20 and 21. We approximate these processes with rapid equilibria as well, which gives

$$\frac{X_{11} X_{12}}{X_{16}} = K_{20} \quad (45)$$

$$\frac{X_5 X_{12}}{X_{17}} = K_{21}. \quad (46)$$

Using these simplifications presented in Eqs. (38)–(46), nine of the 17 differential equations can be eliminated, namely we solve for  $X_1$ ,  $X_5$ ,  $X_6$ ,  $X_8$ ,  $X_{13}$ ,  $X_{14}$ ,  $X_{15}$ ,  $X_{16}$ , and  $X_{17}$ . Several dependent variables remain unchanged from the Lee–Heinrich model. These include the formulas for inactive Dsh ( $X_1$ ), APC/Axin ( $X_6$ ), bound destruction complex ( $X_8$ ), TCF ( $X_{13}$ ), nuclear  $\beta$ -catenin ( $X_{14}$ ), and  $\beta$ -catenin/APC ( $X_{15}$ ), given by

$$X_1 = Dsh^0 - X_2 \quad (47)$$

$$X_6 = \frac{X_7 X_{12}}{K_7} \quad (48)$$

$$X_8 = \frac{X_3 X_{11}}{K_8} \quad (49)$$

$$X_{13} = \frac{K_{16}TCF^0}{K_{16} + X_{11}} \quad (50)$$

$$X_{14} = \frac{X_{11}TCF^0}{K_{16} + X_{11}} \quad (51)$$

$$X_{15} = \frac{X_7 X_{11}}{K_{17}}. \quad (52)$$

The concentration of GSK-3 $\beta$  ( $X_5$ ) is no longer constant, but is eliminated from the system using Eq. (39). Also, we can obtain expressions for the variables  $X_{16}$  and  $X_{17}$ , by rearranging the rapid equilibrium equations above [Eqs. (45), (46)]. We obtain the following dependent variables:

$$X_5 = \frac{K_{21}}{K_{21} + X_{12}} \left[ GSK^0 - \left( 1 + \frac{X_{11}}{K_8} \right) X_3 - X_4 - X_9 \right] \quad (53)$$

$$X_{16} = \frac{X_{11}X_{12}}{K_{20}} \quad (54)$$

$$X_{17} = \frac{X_{12}}{K_{21} + X_{12}} \left[ GSK^0 - \left( 1 + \frac{X_{11}}{K_8} \right) X_3 - X_4 - X_9 \right] \quad (55)$$

We note here the complexity of  $X_5$  and  $X_{17}$  occurs by not assuming that Axin is small. We see a large contribution to the destruction complex concentrations simply due to an increase in all protein complexes that contain Axin. Table 1 gives the remaining eight variables and their description with bolded entries denoting new variables. Using the reaction kinetics given by Lee et al. (in Fig. 1) and the simplifications mentioned above, the governing equations for the remaining variables can be written as

$$\frac{dX_2}{dt} = k_1 W \left( Dsh^0 - X_2 \right) - k_2 X_2 \quad (56)$$

$$\begin{aligned} \frac{dX_4}{dt} = & - (k_3 X_2 + k_4 + k_{-6}) X_4 + k_5 X_3 \\ & + \frac{k_6 K_{21} X_7 X_{12}}{K_7 (K_{21} + X_{12})} \left[ GSK^0 - \left( 1 + \frac{X_{11}}{K_8} \right) X_3 - X_4 - X_9 \right]. \end{aligned} \quad (57)$$

**Table 1** Independent model variables, nondimensional scalings, and initial conditions

| Dimensional variable (nM) | Dimensionless variable | Scaling                      | Notation in Fig. 1           | Initial value      |
|---------------------------|------------------------|------------------------------|------------------------------|--------------------|
| $X_2(t)$                  | $V(\tau)$              | $1/Dsh^0$                    | Dsh <sub>a</sub>             | 0                  |
| $X_3$                     | $D_a$                  | $k_5/k_{14}$                 | APC*/Axin*/GSK               | 15.63              |
| $X_4$                     | $D_i$                  | $k_5/k_{14}$                 | APC/Axin/GSK                 | 7.815              |
| <b><math>X_7</math></b>   | <b><math>P</math></b>  | <b><math>1/K_{17}</math></b> | <b>APC</b>                   | <b>97.9528</b>     |
| $X_9$                     | $D_b$                  | $k_5/k_{14}$                 | $\beta$ -cat*/APC*/Axin*/GSK | 3.268              |
| $X_{10}$                  | $B_p$                  | $k_5 k_{10}/k_{14} k_{11}$   | $\beta$ -cat*                | $7.99 \times 10^5$ |
| $X_{11}$                  | $B_a$                  | $1/K_{16}$                   | $\beta$ -cat                 | 0.836              |
| $X_{12}$                  | $X$                    | $k_5/k_{14}$                 | Axin                         | 0.798              |

$$\frac{dX_9}{dt} = \frac{k_9 X_3 X_{11}}{K_8} - k_{10} X_9 \quad (58)$$

$$\frac{dX_{10}}{dt} = k_{10} X_9 - k_{11} X_{10} \quad (59)$$

$$\left(1 + \frac{X_{11}}{K_8}\right) \frac{dX_3}{dt} + \frac{X_3}{K_8} \frac{dX_{11}}{dt} = k_4 X_4 - \left(k_5 + \frac{k_9 X_{11}}{K_8}\right) X_3 + k_{10} X_9 \quad (60)$$

$$\begin{aligned} \frac{X_{11}}{K_8} \frac{dX_3}{dt} + \frac{X_{11}}{K_{17}} \frac{dX_7}{dt} + \left(1 + \frac{X_3}{K_8} + \frac{K_{16} TCF^0}{(K_{16} + X_{11})^2} + \frac{X_7}{K_{17}} + \frac{X_{12}}{K_{20}}\right) \frac{dX_{11}}{dt} \\ + \frac{X_{11}}{K_{20}} \frac{dX_{12}}{dt} = v_{12} - \left(k_{13} + \frac{k_9 X_3}{K_8}\right) X_{11} \end{aligned} \quad (61)$$

$$\begin{aligned} \left(1 + \frac{X_{11}}{K_8}\right) \frac{dX_3}{dt} + \left(1 + \frac{X_{12}}{K_7} + \frac{X_{11}}{K_{17}}\right) \frac{dX_7}{dt} + \left(\frac{X_3}{K_8} + \frac{X_7}{K_{17}}\right) \frac{dX_{11}}{dt} + \frac{X_7}{K_7} \frac{dX_{12}}{dt} \\ = \frac{v_{18}}{1 + \frac{TCF^0 X_{11}}{K_7(K_{16} + X_{11})} + \frac{X_{11}}{K_b}} - k_{19} X_7 - \left(\frac{dX_4}{dt} + \frac{dX_9}{dt}\right) \end{aligned} \quad (62)$$

$$\begin{aligned} - \frac{X_{12}(K_8 + X_{11})}{K_8(K_{21} + X_{12})} \frac{dX_3}{dt} + \frac{X_{12}}{K_7} \frac{dX_7}{dt} + \left(\frac{X_{12}}{K_{20}} - \frac{X_3 X_{12}}{K_8(K_{21} + X_{12})}\right) \frac{dX_{11}}{dt} \\ + \left\{1 + \frac{X_7}{K_7} + \frac{X_{11}}{K_{20}} + \frac{K_{21} \left[GSK^0 - \left(1 + \frac{X_{11}}{K_8}\right) X_3 - X_4 - X_9\right]}{(K_{21} + X_{12})^2}\right\} \frac{dX_{12}}{dt} \\ = v_{14} + (k_3 X_2 + k_{-6}) X_4 - \frac{k_{15} X_7 X_{12}}{K_m + X_7} + \frac{X_{12}}{K_{21} + X_{12}} \left(\frac{dX_4}{dt} + \frac{dX_9}{dt}\right) \\ - \frac{k_6 K_{21} X_7 X_{12}}{K_7(K_{21} + X_{12})} \left[GSK^0 - \left(1 + \frac{X_{11}}{K_8}\right) X_3 - X_4 - X_9\right]. \end{aligned} \quad (63)$$

Equations (56)–(63) represent the eight equation system that includes the possibility of an increased amount of Axin in the Wnt pathway with APC regulation. Only Eqs. (56), (57), and (58) remained unchanged from the Lee–Heinrich model. We note here that the binding reaction 6 in Eqs. (57) and (63) is different due to the new independent variable  $X_7$ . Equation (63) contains all the new components of the system. We can see how these components influence the change in Axin because there are more binding reactions (e.g., Axin with  $\beta$ -catenin, Axin with GSK- $3\beta$ , and the destruction complex components) included in the system. The strength of these new reactions is measured by the binding constants  $K_{20}$  and  $K_{21}$ , which we assume to be  $\mathcal{O}(1)$  so that both binding reactions have an effect. These components also influence output from Eq. (61), the overall change in  $\beta$ -catenin, and so we should expect an additional limitation of  $\beta$ -catenin accumulation.

We leave Eqs. (60)–(63) in implicit form for ease of nondimensionalization and analysis. To close the system, we consider the same initial conditions as in the Lee–Heinrich paper (Lee et al. 2003), which are also given in Table 1. The additional initial

condition for  $X_7$  is given by the following

$$X_7(0) = \frac{K_{17}APC^0}{K_{17} + X_{11}(0)} = 97.9528, \quad (64)$$

which is an 89% decrease from the initial amount assumed in the Lee–Heinrich model and  $APC^0$  denotes the total amount of APC. This will be our reference value for the amount of APC in the base of the crypt during Wnt stimulation. We note here that Eq. (59) decouples from the system, and will be left out of the discussion for the remainder of the paper.

## 2.2 Nondimensionalization and numerical simulation

We introduce new scaled parameters and nondimensionalize the system so that we can perform a multiple timescale analysis in the next section. We use Mirams' variable scalings given in Table 1, where  $P$  denotes the nondimensional version of  $X_7$ . Also, we scale the time variable  $t$  by the inactivation rate of the destruction complex,  $\tau = k_5 t$ . APC is scaled by  $K_{17}$ , the dissociation constant for APC/ $\beta$ -catenin. The variable for Axin is scaled by the binding constant to GSK-3 $\beta$ ,  $K_{21}$ , so that  $X_{12} = K_{21}X$ . Also, we scale the time variable  $t$  by the inactivation rate of the destruction complex,  $\tau = k_5 t$ . Table 2 (located at the end of the paper) gives all additional nondimensional parameters. Bold entries in Table 1 and in the analysis below denote redefined or new parameters that are different from Mirams et al. Introducing the dimensionless variables and parameters into the remaining equations yields the following nondimensional system (not including the uncoupled equation for phosphorylated  $\beta$ -catenin,  $B_p$ ):

$$\frac{dV}{d\tau} = k'_1 (1 - V) W - k'_2 V \quad (65)$$

$$\frac{dD_b}{d\tau} = k'_9 D_a B_a - k'_{10} D_b \quad (66)$$

$$\begin{aligned} \frac{dD_i}{d\tau} = & -(k'_3 V + k'_4 + k'_{-6}) D_i + D_a \\ & + \frac{k'_6 P X [g s k^0 - (1 + K'_8 B_a) D_a - D_i - D_b]}{1 + X} \end{aligned} \quad (67)$$

$$(1 + K'_8 B_a) \frac{dD_a}{d\tau} + K'_8 D_a \frac{dB_a}{d\tau} = k'_4 D_i - (1 + k'_9 B_a) D_a + k'_{10} D_b \quad (68)$$

$$\begin{aligned} v'_{14b} B_a \frac{dD_a}{d\tau} + B_a \frac{dP}{d\tau} + \left( 1 + v'_{14b} D_a + \frac{t c f^0}{(1 + B_a)^2} + P + K'_{20} X \right) \frac{dB_a}{d\tau} \\ + K'_{20} B_a \frac{dX}{d\tau} = v'_{12} - (k'_{13} + k'_{14} D_a) B_a \end{aligned} \quad (69)$$

$$v'_{14p} (1 + K'_8 B_a) \frac{dD_a}{d\tau} + \left( 1 + K'_{21} X + K'_{16p} B_a \right) \frac{dP}{d\tau} + K'_{16p} (v'_{14b} D_a + P) \frac{dB_a}{d\tau}$$

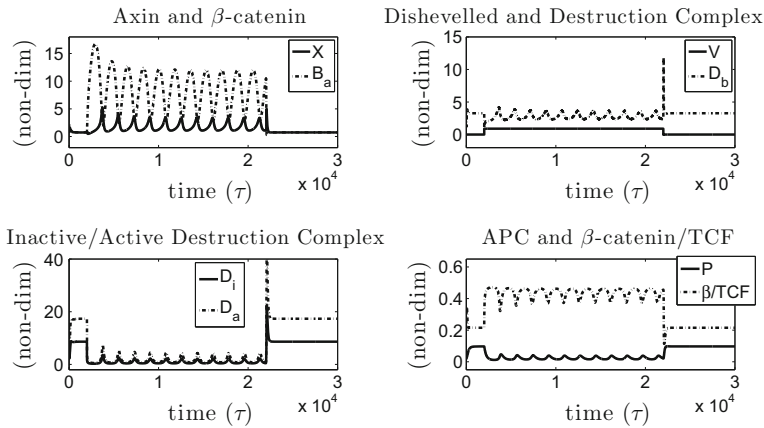
**Table 2** Dimensional and dimensionless parameters and estimates for APC and Axin regulation model of Sect. 3

| Parameter (dim) | Estimate (dim)        | Units                            | Parameter (non-dim)        | Estimate (non-dim) | Scaled Parameter (non-dim)                    | Estimate (non-dim) |
|-----------------|-----------------------|----------------------------------|----------------------------|--------------------|-----------------------------------------------|--------------------|
| $Dsh^0$         | 100.0                 | nM                               | —                          | —                  | —                                             | —                  |
| $APC^0$         | 100.0                 | nM                               | —                          | —                  | —                                             | —                  |
| $TCF^0$         | 15.0                  | nM                               | $tc f^0 = TCF^0/K_{16}$    | 0.50               | $tc f^0 = \varepsilon^0 tc f^0$               | 0.50               |
| $GSK^0$         | 50.0                  | nM                               | $gsk^0 = GSK^0 k_5/v_{14}$ | $8.09 \times 10^4$ | $gsk^0 = \varepsilon^0 gsk^0$                 | $8.09 \times 10^4$ |
| $K_7$           | 50.0                  | nM                               | $K'_7 = K_{17}/K_7$        | 24.0               | $K'_7 = \varepsilon^0 \widehat{K'_7}$         | 24.0               |
| $K_8$           | 120.0                 | nM                               | $K'_8 = K_{16}/K_8$        | 0.25               | $K'_8 = \varepsilon^0 \widehat{K'_8}$         | 0.25               |
| $K_{16}$        | 30.0                  | nM                               | $K'_{16a} = K_{16}/K_{21}$ | 30.0               | $K'_{16a} = \varepsilon^0 \widehat{K'_{16a}}$ | 30.0               |
|                 |                       |                                  | $K'_{16p} = K_{16}/K_{17}$ | 0.025              | $K'_{16p} = \varepsilon^1 \widehat{K'_{16p}}$ | 12.95              |
| $K_{17}$        | 1200                  | nM                               | —                          | —                  | —                                             | —                  |
| $K_{20}$        | 1                     | nM                               | $K'_{20} = K_{21}/K_{20}$  | 1                  | $K'_{20} = \varepsilon^0 \widehat{K'_{20}}$   | 1                  |
| $K_{21}$        | 1                     | nM                               | $K'_{21} = K_{21}/K_7$     | 0.02               | $K'_{21} = \varepsilon \widehat{K'_{21}}$     | 10.36              |
| $K_t$           | 39.2115               | nM                               | $K'_t = TCF^0/K_t$         | 0.3825             | $K'_t = \varepsilon^0 \widehat{K'_t}$         | 0.3825             |
| $K_b$           | 34.0445               | nM                               | $K'_b = K_{16}/K_b$        | 0.8815             | $K'_b = \varepsilon^0 \widehat{K'_b}$         | 0.8815             |
| $K_m$           | 98                    | nM                               | $K'_m = K_m/K_{17}$        | 0.0817             | $K'_m = \varepsilon^0 \widehat{K'_m}$         | 0.0817             |
| $k_1$           | 0.182                 | $\text{min}^{-1}$                | $k'_1 = k_1/k_5$           | 1.37               | $k'_1 = \varepsilon^0 \widehat{k'_1}$         | 1.37               |
| $k_2$           | $1.82 \times 10^{-2}$ | $\text{min}^{-1}$                | $k'_2 = k_2/k_5$           | 0.137              | $k'_2 = \varepsilon^0 \widehat{k'_2}$         | 0.137              |
| $k_3$           | $5.0 \times 10^{-2}$  | $\text{nM}^{-1} \text{min}^{-1}$ | $k'_3 = Dsh^0 k_3/k_5$     | 37.6               | $k'_3 = \varepsilon^0 \widehat{k'_3}$         | 0.376              |
| $k_4$           | 0.267                 | $\text{min}^{-1}$                | $k'_4 = k_4/k_5$           | 2.01               | $k'_4 = \varepsilon^0 \widehat{k'_4}$         | 2.01               |

**Table 2** continued

| Parameter (dim) | Estimate (dim)        | Units                            | Parameter (non-dim)                  | Estimate (non-dim)     | Scaled Parameter (non-dim)                     | Estimate (non-dim) |
|-----------------|-----------------------|----------------------------------|--------------------------------------|------------------------|------------------------------------------------|--------------------|
| $k_5$           | 0.133                 | $\text{min}^{-1}$                | —                                    | —                      | —                                              | —                  |
| $k_6$           | $9.09 \times 10^{-2}$ | $\text{nM}^{-1} \text{min}^{-1}$ | $k'_6 = k_6 K_{17} K_{21} / k_5 K_7$ | <b>16.403</b>          | $k'_6 = \varepsilon^0 \widehat{k'_6}$          | <b>16.403</b>      |
| $k_{-6}$        | 0.909                 | $\text{min}^{-1}$                | $k'_{-6} = k_{-6} / k_5$             | 6.84                   | $k'_{-6} = \varepsilon^0 \widehat{k'_{-6}}$    | 6.84               |
| $k_9$           | 206.0                 | $\text{min}^{-1}$                | $k'_9 = k_9 K_{16} / k_5 K_8$        | 387.25                 | $k'_9 = \varepsilon^{-1} \widehat{k'_9}$       | 0.75               |
| $k_{10}$        | 206.0                 | $\text{min}^{-1}$                | $k'_{10} = k_{10} / k_5$             | 1549                   | $k'_{10} = \varepsilon^{-1} \widehat{k'_{10}}$ | 2.99               |
|                 |                       |                                  | $v'_{10} = k_{10} k'_{10} / k_{11}$  | $7.65 \times 10^5$     | —                                              | —                  |
| $k_{11}$        | 0.417                 | $\text{min}^{-1}$                | $k'_{11} = k_{11} / k_5$             | 3.14                   | $k'_{11} = \varepsilon^0 \widehat{k'_{11}}$    | 3.14               |
| $v_{12}$        | 0.423                 | $\text{nM min}^{-1}$             | $v'_{12} = v_{12} / k_5 K_{16}$      | 0.106                  | $v'_{12} = \varepsilon^1 \widehat{v'_{12}}$    | 54.92              |
| $k_{13}$        | $2.57 \times 10^{-4}$ | $\text{min}^{-1}$                | $k'_{13} = k_{13} / k_5$             | $1.93 \times 10^{-3}$  | $\varepsilon = k'_{13}$                        | —                  |
| $v_{14}$        | $8.22 \times 10^{-5}$ | $\text{nM min}^{-1}$             | $v'_{14a} = v_{14} / k_5 K_{21}$     | $6.18 \times 10^{-4}$  | $v'_{14a} = \varepsilon^1 \widehat{v'_{14a}}$  | <b>0.320</b>       |
|                 |                       |                                  | $v'_{14b} = v_{14} / k_5 K_8$        | $5.15 \times 10^{-6}$  | $v'_{14b} = \varepsilon^2 \widehat{v'_{14b}}$  | <b>1.382</b>       |
|                 |                       |                                  | $v'_{14p} = v_{14} / k_5 K_{16}$     | $5.15 \times 10^{-7}$  | $v'_{14p} = \varepsilon^2 \widehat{v'_{14p}}$  | <b>0.138</b>       |
|                 |                       |                                  | $k'_{14} = k_9 v'_{14b} / k_5$       | $7.98 \times 10^{-3}$  | $k'_{14} = \varepsilon^1 \widehat{k'_{14}}$    | <b>4.13</b>        |
| $k_{15}$        | <b>0.333</b>          | $\text{min}^{-1}$                | $k'_{15} = k_{15} / k_5$             | <b>2.498</b>           | $k'_{15} = \varepsilon^1 \widehat{k'_{15}}$    | $1.29 \times 10^3$ |
| $v_{18}$        | 0.1774                | $\text{nM min}^{-1}$             | $v'_{18} = v_{18} / k_5 K_{17}$      | $1.111 \times 10^{-3}$ | $v'_{18} = \varepsilon^1 \widehat{v'_{18}}$    | 0.576              |
| $k_{19}$        | $8.33 \times 10^{-4}$ | $\text{min}^{-1}$                | $k'_{19} = k_{19} / k_5$             | $6.265 \times 10^{-3}$ | $k'_{19} = \varepsilon^1 \widehat{k'_{19}}$    | 3.246              |
|                 |                       |                                  | $W$                                  | 1.00                   | $W = \varepsilon^0 \widehat{W}$                | 1.00               |

All parameters are fit or derived from experiment by [Lee et al. \(2003\)](#). Bolded entries constitute new parameters



**Fig. 2** A simulation of Eqs. (65)–(70). (Top left) Axin (solid) and free  $\beta$ -catenin (dashed). Axin is scaled by  $v_{14}/k_5$  for presentation. (Top right) active Dsh (solid) and bound destruction complex (dashed). (Bottom left) inactive destruction complex (solid) and active destruction complex (dashed). (Bottom right) free APC (solid) and nuclear  $\beta$ -catenin,  $\beta$ -catenin/TCF (dashed)

$$\begin{aligned}
 & + K'_{21} P \frac{dX}{d\tau} = \frac{v'_{18}}{1 + \frac{K'_1 B_a}{1+B_a} + K'_b B_a} - k'_{19} P - v'_{14p} \left( \frac{dD_i}{d\tau} + \frac{dD_b}{d\tau} \right) \quad (70) \\
 & - v'_{14a} \frac{(1 + K'_8 B_a) X}{1 + X} \frac{dD_a}{d\tau} + K'_7 X \frac{dP}{d\tau} + K'_{16a} \left( K'_{20} X - v'_{14b} \frac{D_a X}{1 + X} \right) \frac{dB_a}{d\tau} \\
 & + \left\{ 1 + K'_7 P + K'_{16a} K'_{20} B_a + \frac{v'_{14a} [gsk^0 - (1 + K'_8 B_a) D_a - D_i - D_b]}{(1 + X)^2} \right\} \frac{dX}{d\tau} \\
 & = v'_{14a} + v'_{14a} (k'_3 V + k'_{-6}) D_i - \frac{k'_{15} P X}{K'_m + P} + v'_{14a} \frac{X}{1 + X} \left( \frac{dD_i}{d\tau} + \frac{dD_b}{d\tau} \right) \\
 & - v'_{14a} \frac{k'_6 P X [gsk^0 - (1 + K'_8 B_a) D_a - D_i - D_b]}{1 + X}. \quad (71)
 \end{aligned}$$

Only Eqs. (65) and (66) remained unchanged from the analysis of Mirams et al. (2010). Here, Eqs. (68)–(71) are again presented implicitly for simplification. To solve this system numerically, we use the original initial conditions and use the stiff ODE solver `ode15s` in MATLAB. We present typical results showing the system's response to the following Wnt signal given by

$$W = \begin{cases} 1 & 2000 \leq \tau \leq 22000 \\ 0 & \text{otherwise} \end{cases} \quad (72)$$

in Fig. 2. Here, the Wnt signal is established at  $\tau = 2000$  dimensionless time units, but remains on much longer until  $\tau = 22000$  time units so that we can now observe the oscillatory behavior of the system occurring at large time. All parameters used for the simulation are presented in Table 2. It is evident from the figure that every

nondimensional variable achieves a new quasi-steady-state before the Wnt signal is induced. Furthermore, in the presence of Wnt, we observe oscillatory behavior of all system variables except activated Dsh, which is unsurprising since Dsh is only altered by the Wnt signal. This is due to the dual feedback loops between  $\beta$ -catenin and APC, and APC and Axin. That is, when  $\beta$ -catenin becomes prominent in the cell during Wnt stimulation, the amount of APC goes down because the cell is essentially very proliferative, which regulates the amount of APC through the additional feedback. When APC diminishes, however, the amount of Axin increases since its degradation is antagonized. Hence, the slight increase of Axin yields more active destruction complex activity; thereby, limiting the accumulation of  $\beta$ -catenin. Finally, when  $\beta$ -catenin is lacking, the cell is assumed to be in a more differentiated state, which increases the levels of APC, which in effect drives down the amount of Axin and  $\beta$ -catenin, creating the cycle. In essence, there are three regulatory mechanisms driving the oscillatory behavior: the inverse expression of APC and  $\beta$ -catenin, diminished APC upregulating Axin, and the function of APC and Axin down-regulating  $\beta$ -catenin via Wnt transduction.

Further, the protein complex concentrations containing  $\beta$ -catenin, Axin, and/or APC oscillate as well. All three forms of the destruction complex oscillate since they include both APC and Axin. The nuclear protein complex consisting of  $\beta$ -catenin and TCF oscillates as well. This suggests that the core components of the system are  $\beta$ -catenin, Axin, and APC, and we expect the following multiple timescale analysis to show this fact. It can also be observed that all levels of participating proteins switch back to the original state when the Wnt stimulation is turned off. This unstimulated state is stable, which indicates a Hopf bifurcation may exist with the Wnt signal,  $W$ , likely to be a control parameter.

Several biological studies have concluded that  $\beta$ -catenin levels are observed to oscillate in many systems (Lloyd-Lewis et al. 2013; Orford et al. 1997; Cliffe et al. 2003). For instance, the Wnt target gene *Axin2*, which was mathematically modeled by Wawra et al. (2007), is known to have a pro-proliferative role at the G2/M phase of cell mitosis (Hadjihannas et al. 2003), as well as regulate the amount of the protein Axin, which as we know, down-regulates  $\beta$ -catenin through binding of the destruction complex. This suggests that the cell may move between two different proliferative states as the Wnt pathway operates. Furthermore, as hypothesized by Boman and Fields (2013), a fluctuation in Wnt factors, namely through an increase of APC, at the base of the crypt may be responsible for initiation of crypt fission. In the present model, Axin concentrations are driven by a diminishing APC expression, which is affected by the state of both free and nuclear  $\beta$ -catenin. Hence, a cell's ability to switch between a proliferating or quiescent state is completely determined by the Wnt signal and how it regulates the amounts of  $\beta$ -catenin, APC, and Axin.

Lastly, the added reactions help to limit the increase in  $\beta$ -catenin concentration, as expected. During Wnt stimulation, Lee's original model shows about a fivefold increase in free  $\beta$ -catenin. APC regulation coupled with the Axin degradation yields oscillatory  $\beta$ -catenin levels, but solving for the equilibrium value yields an approximate eightfold increase in  $\beta$ -catenin. Therefore, our model does well as it restricts the levels of  $\beta$ -catenin to a realistic value, while incorporating both the inverse relationship with APC and the recruitment function of Axin.

### 2.3 Multiple timescale analysis

We next perform a multiple timescale analysis as in the paper by [Mirams et al. \(2010\)](#). Because there exists a dynamic interplay between APC, Axin, and  $\beta$ -catenin, we expect the analysis to return three equations on the long timescale. Hence, most of the simplification will be a rapid equilibrium approximation to the formation and activation of the destruction complex, and a long-term expression for Dsh. In essence, we will simplify the system given by Eqs. (65)–(71) to a system of three equations that capture similar long-term dynamics of the full system.

We introduce the small parameter  $\varepsilon = k'_{13} = k_{13}/k_5$ , which is a measure of the rate of independent destruction of  $\beta$ -catenin to the rate at which the destruction complex is activated. All scalings are provided in Table 2 (located at the end of the paper). For this model and the remainder of the section, we use APC regulation parameters derived to fit an estimated APC gradient of the colonic crypt. Although the mechanism of APC synthesis is not well known, we define the dimensionless parameters  $v'_{18} = \varepsilon \widehat{v'_{18}}$  and  $k'_{19} = \varepsilon \widehat{k'_{19}}$ , so that APC synthesis works on the same timescale as the degradation of  $\beta$ -catenin by the destruction complex. Therefore, the change in APC is determined when the Wnt pathway influences the level of  $\beta$ -catenin. Furthermore, we want the model to adapt to an increase in total Axin concentration. This means we'll have to be selective in our choice of scaled parameters—some mathematically “small” terms (as presented in the current parameter set) may be included in long-term dynamics. For instance, the rate of Axin degradation,  $k_{15}$ , operates on the same scale of  $\beta$ -catenin and APC dynamics and so it will receive the appropriate scaling,  $\widehat{k'_{15}} = \varepsilon^{-1} k'_{15}$ . Also, the destruction complex may be of the same magnitude as GSK-3 $\beta$ , and hence the equations involving the term

$$gsk^0 - (1 + K'_8 B_a) D_a - D_i - D_b \quad (73)$$

are handled appropriately. Upon substitution of the scaled parameters into our seven equations, we obtain the following system:

$$\frac{dV}{d\tau} = \widehat{k'_1}(1 - V)W - \widehat{k'_2}V \quad (74)$$

$$\varepsilon \frac{dD_b}{d\tau} = \widehat{k'_9} D_a B_a - \widehat{k'_{10}} D_b \quad (75)$$

$$\begin{aligned} \frac{dD_i}{d\tau} = & -(\widehat{k'_3}V + \widehat{k'_4} + \widehat{k'_{-6}}) D_i + D_a \\ & + \frac{\widehat{k'_6} P X \left[ gsk^0 - (1 + \widehat{K'_8} B_a) D_a - D_i - D_b \right]}{1 + X} \end{aligned} \quad (76)$$

$$(1 + \widehat{K'_8} B_a) \frac{dD_a}{d\tau} + \widehat{K'_8} D_a \frac{dB_a}{d\tau} = \widehat{k'_4} D_i - \left( 1 + \frac{1}{\varepsilon} \widehat{k'_9} B_a \right) D_a + \frac{1}{\varepsilon} \widehat{k'_{10}} D_b \quad (77)$$

$$\varepsilon \widehat{v'_{14b}} B_a \frac{dD_a}{d\tau} + \frac{1}{\varepsilon} B_a \frac{dP}{d\tau} + \left( \frac{1}{\varepsilon} + \widehat{v'_{14b}} D_a + \frac{1}{\varepsilon} \frac{tcf^0}{(1 + B_a)^2} + \frac{1}{\varepsilon} P + \frac{1}{\varepsilon} \widehat{K'_{20}} X \right) \frac{dB_a}{d\tau}$$

$$+ \frac{1}{\varepsilon} \widehat{K'_{20}} B_a \frac{dX}{d\tau} = \widehat{v'_{12}} - \left(1 + \widehat{k'_{14}} D_a\right) B_a \quad (78)$$

$$\begin{aligned} & \varepsilon \widehat{v'_{14p}} \left(1 + \widehat{K'_8} B_a\right) \frac{dD_a}{d\tau} + \left(\frac{1}{\varepsilon} + \widehat{K'_{21}} X + \widehat{K'_{16p}} B_a\right) \frac{dP}{d\tau} \\ & + \widehat{K'_{16p}} \left(\varepsilon^2 \widehat{v'_{14b}} D_a + P\right) \frac{dB_a}{d\tau} + \widehat{K'_{21}} P \frac{dX}{d\tau} = \frac{\widehat{v'_{18}}}{1 + \frac{\widehat{K'_8} B_a}{1+B_a} + \widehat{K'_b} B_a} \\ & - \widehat{k'_{19}} P - \varepsilon \widehat{v'_{14p}} \left(\frac{dD_i}{d\tau} + \frac{dD_b}{d\tau}\right) \end{aligned} \quad (79)$$

$$\begin{aligned} & - \widehat{v'_{14a}} \frac{\left(1 + \widehat{K'_8} B_a\right) X}{1+X} \frac{dD_a}{d\tau} + \frac{1}{\varepsilon} \widehat{K'_7} X \frac{dP}{d\tau} + \frac{1}{\varepsilon} \widehat{K'_{16a}} \left(\widehat{K'_{20}} X - \varepsilon^2 \widehat{v'_{14b}} \frac{D_a X}{1+X}\right) \frac{dB_a}{d\tau} \\ & + \frac{1}{\varepsilon} \left\{ 1 + \widehat{K'_7} P + \widehat{K'_{16a}} \widehat{K'_{20}} B_a \right. \\ & \left. + \varepsilon \frac{\widehat{v'_{14a}} \left[ \widehat{gsk^0} - \left(1 + \widehat{K'_8} B_a\right) D_a - D_i - D_b \right]}{(1+X)^2} \right\} \frac{dX}{d\tau} \\ & = \widehat{v'_{14a}} + \widehat{v'_{14a}} \left(\widehat{k'_3} V + \widehat{k'_{-6}}\right) D_i - \frac{\widehat{k'_{15}} P X}{\widehat{K'_m} + P} + \widehat{v'_{14a}} \frac{X}{1+X} \left(\frac{dD_i}{d\tau} + \frac{dD_b}{d\tau}\right) \\ & - \widehat{v'_{14a}} \frac{\widehat{k'_6} P X \left[ \widehat{gsk^0} - \left(1 + \widehat{K'_8} B_a\right) D_a - D_i - D_b \right]}{1+X}. \end{aligned} \quad (80)$$

It is evident from these equations that the system works on three timescales: short timescale,  $\mathcal{O}(\varepsilon)$ ; intermediate timescale,  $\mathcal{O}(1)$ ; and long timescale,  $\mathcal{O}(\varepsilon^{-1})$ . We investigate each timescale using a formal asymptotic approach by substituting a power series expansion in  $\varepsilon$  for each independent variable.

Introducing the short timescale variable  $t = \varepsilon^{-1} \tau$ , the nondimensional approximation for concentration of  $\beta$ -catenin,  $B_a(t)$ , is represented as the expansion

$$B_a(t) = B_{a0}(t) + \varepsilon B_{a1}(t) + \varepsilon^2 B_{a2}(t) + \cdots,$$

where  $B_{a0}$  is the zeroth (leading) order approximation,  $B_{a1}$  is the first-order approximation, and so on. All other nondimensional variables are defined with a similar expansion. In this case, a system emerges in which only the active and bound phases of the destruction complex evolve with time and all other variables remain fixed at the initial conditions. We obtain the following linear system at leading order for  $D_a$  and  $D_b$ :

$$\frac{d}{dt} \begin{bmatrix} D_{a0} \\ D_{b0} \end{bmatrix} = \begin{bmatrix} -\frac{\widehat{k'_9} B_a(0)}{1 + \widehat{K'_8} B_a(0)} & \frac{\widehat{k'_{10}}}{1 + \widehat{K'_8} B_a(0)} \\ \widehat{k'_9} B_a(0) & -\widehat{k'_{10}} \end{bmatrix} \begin{bmatrix} D_{a0} \\ D_{b0} \end{bmatrix} \quad (81)$$

where  $B_a(0)$  is the initial value of  $\beta$ -catenin. This timescale is characterized by the phosphorylation of bound  $\beta$ -catenin by the active destruction complex. Here, we can

see that the rows of our matrix are linearly dependent, yielding the following eigenvalues

$$\lambda_0 = 0, \quad \lambda_1 = -\left(\widehat{k'_{10}} + \frac{B_a(0)}{1 + \widehat{K'_8} B_a(0)} \widehat{k'_9}\right), \quad (82)$$

indicating that the system is stable. We obtain solutions of the following form

$$D_{a0}(t) = a_0 + a_1 e^{\lambda_1 t} \quad (83)$$

$$D_{b0}(t) = b_0 + b_1 e^{\lambda_1 t}, \quad (84)$$

where

$$a_0 = \frac{D_a(0) B_a(0) \widehat{k'_9} - D_b(0) \widehat{k'_{10}}}{B_a(0) \widehat{k'_9} + (1 + \widehat{K'_8} B_a(0)) \widehat{k'_{10}}} \quad (85)$$

$$a_1 = \frac{\left[(1 + \widehat{K'_8} B_a(0)) D_a(0) + D_b(0)\right] \widehat{k'_{10}}}{B_a(0) \widehat{k'_9} + (1 + \widehat{K'_8} B_a(0)) \widehat{k'_{10}}} \quad (86)$$

$$b_0 = \frac{-(1 + \widehat{K'_8} B_a(0)) D_a(0) B_a(0) \widehat{k'_9} + (1 + \widehat{K'_8} B_a(0)) D_b(0) \widehat{k'_{10}}}{B_a(0) \widehat{k'_9} + (1 + \widehat{K'_8} B_a(0)) \widehat{k'_{10}}} \quad (87)$$

$$b_1 = \frac{\left[(1 + \widehat{K'_8} B_a(0)) D_a(0) + D_b(0)\right] B_a(0) \widehat{k'_9}}{B_a(0) \widehat{k'_9} + (1 + \widehat{K'_8} B_a(0)) \widehat{k'_{10}}}. \quad (88)$$

Here  $D_a(0)$  and  $D_b(0)$  are the initial concentrations. Therefore, both  $D_{a0}$  and  $D_{b0}$  decay to a constant value on this timescale, and as  $\tau$  surpasses  $\mathcal{O}(\varepsilon^{-1})$ , this approximation is no longer valid.

On the intermediate timescale, we introduce the variable  $\tau = \varepsilon^0 \tau$  and we seek solutions as a power series expansion in  $\varepsilon$ . Upon substitution into the scaled equations, we find that  $\beta$ -catenin, Axin, and APC remain constant to leading order and the concentration of the bound destruction complex is approximated by a rapid equilibrium between free  $\beta$ -catenin and the active destruction complex:

$$D_{b0}(\tau) = \frac{\widehat{k'_9}}{\widehat{k'_{10}}} D_{a0} B_{a0}. \quad (89)$$

All remaining variables evolve over time according to the following three equations:

$$\frac{dV_0}{d\tau} = \widehat{k'_1} (1 - V_0) W - \widehat{k'_2} V_0 \quad (90)$$

$$\begin{aligned} \frac{dD_{i0}}{d\tau} = & -\left(\widehat{k'_3} V_0 + \widehat{k'_4} + \widehat{k'_{-6}}\right) D_{i0} + D_{a0} \\ & + \frac{\widehat{k'_6} P_0 X_0}{1 + X_0} \left[\widehat{g s k^0} - \left(1 + \widehat{K'_8} B_{a0}\right) D_{a0} - D_{i0} - D_{b0}\right] \end{aligned} \quad (91)$$

$$\frac{dD_{a0}}{d\tau} = \frac{\widehat{k'_4} D_{i0} - D_{a0}}{1 + \left( \widehat{K'_8} + \frac{\widehat{k'_9}}{\widehat{k'_{10}}} \right) B_{a0}}. \quad (92)$$

Making the proper substitutions, we can solve explicitly for active Dishevelled at leading order:

$$V_0(\tau) = \frac{k'_1 W}{k'_1 W + k'_2} + \left( V(0) - \frac{k'_1 W}{k'_1 W + k'_2} \right) e^{-(k'_1 W + k'_2)\tau}. \quad (93)$$

This leaves a  $2 \times 2$  system in the active and inactive destruction complexes. In this timescale, the Wnt signal has affected most of the system similar to the [Mirams et al. \(2010\)](#) analysis. In contrast to the original model, Axin stays constant to leading order since both APC and  $\beta$ -catenin influence its degradation. We see that activation of the destruction complex is the characteristic activity on the intermediate timescale.

For the long-term dynamics, we let  $T = \varepsilon\tau$  to obtain the following set of constant solutions at leading order:

$$V_0(T) = \frac{\widehat{k'_1} \widehat{W}}{\widehat{k'_1} \widehat{W} + \widehat{k'_2}} \quad (94)$$

$$D_{b0}(T) = \frac{\widehat{k'_9}}{\widehat{k'_{10}}} D_{a0} B_{a0} \quad (95)$$

$$D_{i0}(T) = \frac{\widehat{k'_6} \widehat{g} \widehat{s} \widehat{k^0} P_0 X_0}{\left( \widehat{k'_3} V_0 + \widehat{k'_{-6}} \right) (1 + X_0) + \widehat{k'_6} P_0 X_0 \left\{ 1 + \left[ 1 + \left( \widehat{K'_8} + \frac{\widehat{k'_9}}{\widehat{k'_{10}}} \right) B_{a0} \right] \widehat{k'_4} \right\}} \quad (96)$$

$$D_{a0}(T) = \widehat{k'_4} D_{i0} \quad (97)$$

We are left with the following differential equations that describe the evolution of  $\beta$ -catenin, APC, and Axin at leading order:

$$B_{a0} \frac{dP_0}{dT} + \left( 1 + P_0 + \widehat{K'_{20}} X_0 \right) \frac{dB_{a0}}{dT} + \widehat{K'_{20}} B_{a0} \frac{dX_0}{dT} = \widehat{v'_{12}} - \left( 1 + \widehat{k'_{14}} D_{a0} \right) B_{a0} \quad (98)$$

$$\frac{dP_0}{dT} = \frac{\widehat{v'_{18}}}{1 + \frac{\widehat{K'_t} B_{a0}}{1 + B_{a0}} + \widehat{K'_b} B_{a0}} - \widehat{k'_{19}} P_0 \quad (99)$$

$$\widehat{K'_7} X_0 \frac{dP_0}{dT} + \widehat{K'_{16a}} \widehat{K'_{20}} X_0 \frac{dB_{a0}}{dT} + \left\{ 1 + \widehat{K'_7} P_0 + \widehat{K'_{16a}} \widehat{K'_{20}} B_{a0} \right\}$$

$$\begin{aligned}
& + \varepsilon \widehat{v'_{14a}} \frac{a(D_{i0}, B_{a0})}{1 + X_0} \left\} \frac{dX_0}{dT} = \widehat{v'_{14a}} + \widehat{v'_{14a}} \left( \widehat{k'_3} V_0 + \widehat{k'_{-6}} \right) D_{i0} \\
& - \widehat{v'_{14a}} \widehat{k'_6} a(D_{i0}, B_{a0}) P_0 X_0 - \frac{\widehat{k'_{15}} P_0 X_0}{\widehat{K'_m} + P_0}, \tag{100}
\end{aligned}$$

where

$$a(D_{i0}, B_{a0}) = \frac{\widehat{gsk^0} - \left\{ 1 + \left[ 1 + \left( \widehat{K'_8} + \frac{\widehat{k'_9}}{\widehat{k'_{10}}} \right) B_{a0} \right] \widehat{k'_4} \right\} D_{i0}}{1 + X_0} \tag{101}$$

We note the difference in Eq. (96), the term that describes the quantity of inactive destruction complex ( $D_i$ ), from the analysis of Mirams et al. (2010). The denominator contains additional terms associated with the formation of the destruction complex that were assumed to be small before. This makes the remaining three differential equations more complex. The only explicit equation is that of APC regulation, given by Eq. (99). The equations for unphosphorylated  $\beta$ -catenin and Axin are given by Eqs. (98) and (100), respectively, and are presented implicitly.

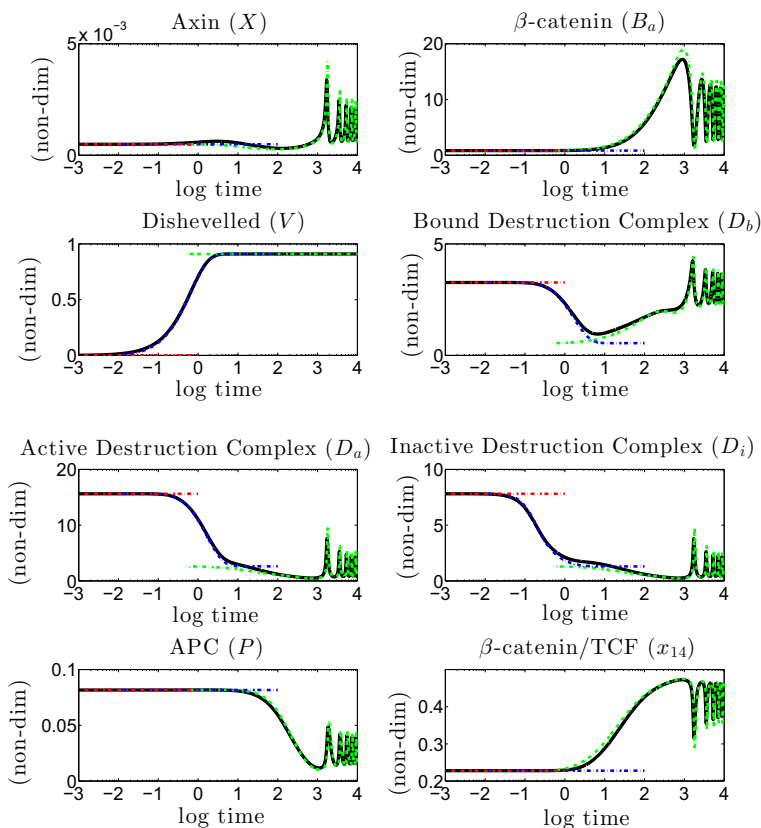
We include the term  $\varepsilon \widehat{v'_{14a}} a(D_{i0}, B_{a0}) / (1 + X_0)$  in the analysis for two reasons: without it, an accurate approximation fails; and the term  $a(D_{i0}, B_{a0})$  balances  $\varepsilon$  since  $\widehat{gsk^0}$  is large, and we do not want to discount the additional terms included in  $a(D_{i0}, B_{a0})$  as they could be of comparable size to  $\widehat{gsk^0}$  in the presence of large Axin. Hence, this term is essentially of first order. Since  $\widehat{gsk^0}$  is large, the term  $\widehat{v'_{14a}} \widehat{k'_6} a(D_{i0}, B_{a0}) P_0 X_0$ , which appears later in the equation, must also be large. We argue that this term is also  $\mathcal{O}(1)$  because it is being multiplied by  $X_0$ , which from the original data, is very small,  $X_0 \sim 0$ . Therefore, Eq. (100) shows the influence of a larger leading order term in the Axin expansion, which is ultimately determined through measuring the total amount of Axin or the flux of Axin through the system, i.e.,  $v_{14}$  and  $k_{15}$ . These terms are essential in describing the added contribution of Axin into the system. Ultimately, the degradation of  $\beta$ -catenin depends on the change in Axin and APC.

Figure 3 shows the combination of the three timescale approximations compared to the solution using the full system, given by Eqs. (74)–(80). The plot is presented on a log-time axis, which demonstrates the time at which each approximation is valid. We note here that the amount of Axin is smaller on the short timescale since it is scaled differently in this model. Furthermore, we note that Axin stays nearly constant at both the short and intermediate timescales until it is influenced by the change in  $\beta$ -catenin and APC for  $\tau \sim \mathcal{O}(\varepsilon^{-1})$ . We can see that each timescale approximation is in good agreement with the full system, and we can effectively construct a composite solution using each of the three approximations.

The long-term dynamics of the full model are presented explicitly in the original parameters below:

$$\frac{dX_7}{dt} = F \tag{102}$$

$$\frac{dX_{11}}{dt} = \frac{\left( 1 + \frac{X_7}{K_7} + \frac{X_{11}}{K_{20}} + \frac{K_{21}}{(K_{21} + X_{12})g} \right) G - \frac{X_{11}}{K_{20}} H}{\left( 1 + \frac{X_7}{K_7} + \frac{X_{12}}{K_{20}} \right) \left( 1 + \frac{X_7}{K_7} + \frac{X_{11}}{K_{20}} + \frac{K_{21}}{(K_{21} + X_{12})g} \right) - \frac{X_{11} X_{12}}{K_{20}^2}} \tag{103}$$



**Fig. 3** A semi-log plot showing the nondimensional solutions for  $X$ ,  $B_a$ ,  $V$ ,  $D_b$ ,  $D_a$ ,  $D_i$ ,  $P$ , and  $x_{14}$  (non-dimensional  $\beta$ -catenin/TCF) on each timescale for  $W = 1$ . The black solid line represents the solution to the full system [Eqs. (74)–(79)]. Short [Eq. (81)], intermediate [Eqs. (89), (90)–(92)], and long [Eqs. (94)–(99)] time approximations are given by the red, blue, and green dashed-dotted lines, respectively. Parameter values are in Table 2 (color figure online)

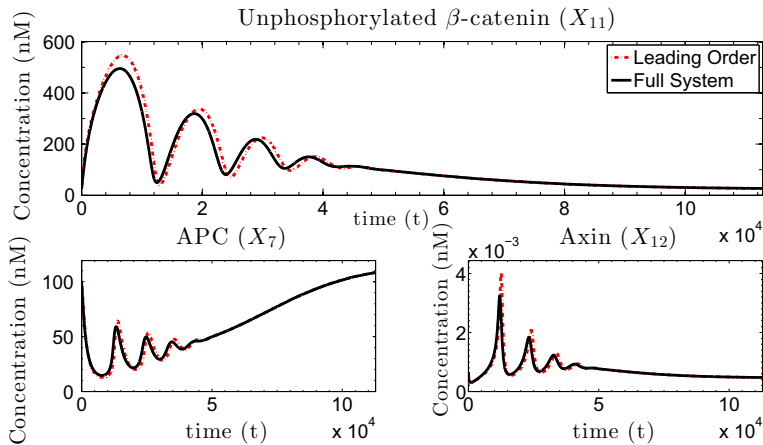
$$\frac{dX_{12}}{dt} = \frac{-\frac{X_{12}}{K_{20}}G + \left(1 + \frac{X_7}{K_7} + \frac{X_{12}}{K_{20}}\right)H}{\left(1 + \frac{X_7}{K_7} + \frac{X_{12}}{K_{20}}\right)\left(1 + \frac{X_7}{K_7} + \frac{X_{11}}{K_{20}} + \frac{K_{21}}{(K_{21} + X_{12})g}\right) - \frac{X_{11}X_{12}}{K_{20}^2}} \quad (104)$$

where

$$F = \frac{v_{18}}{1 + \frac{TCF^0 X_{11}}{K_i(K_{16} + X_{11})} + \frac{X_{11}}{K_b}} - k_{19}X_7 \quad (105)$$

$$G = v_{12} - \left(k_{13} + X_7 X_{12} \frac{f}{g} + \frac{F}{K_{17}}\right) X_{11} \quad (106)$$

$$H = v_{14} - \left(\frac{k_{15}X_7}{K_m + X_7} + \frac{F}{K_7}\right) X_{12}, \quad (107)$$



**Fig. 4** A comparison of the full system of nondimensional equations and the leading order solution with a transient Wnt signal given by  $W = e^{-t/2500}$  and  $v_{18} = 0.1774$ ,  $k_{19} = 8.33 \times 10^{-4}$ ,  $K_t = 39.2115$ ,  $K_b = 34.0335$

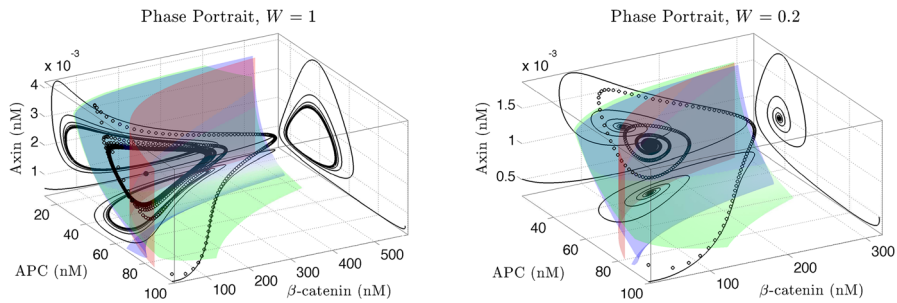
and the functions  $f$  and  $g$  are given by

$$f = \frac{k_1 k_4 k_6 k_9 K_{21}}{k_5 K_7 K_8} \left( \frac{W + k_2/k_1}{k_1 (Dsh^0 k_3 + k_{-6}) W + k_2 k_{-6}} \right) \quad (108)$$

$$g = \frac{K_{21} + X_{12}}{GSK^0} + \frac{(k_9 + k_{10}) X_7 X_{12}}{k_9 k_{10} GSK^0} \left( \frac{(k_4 + k_5) K_8 k_{10}}{k_4 (k_9 + k_{10})} + X_{11} \right) f. \quad (109)$$

We can see that there are four types of  $\beta$ -catenin degradation: a free form destruction ( $k_{13}$ ), a Wnt and APC mediated degradation by the destruction complex ( $f(W)$ ), a decrease by the change in APC ( $F$ ), and an additional degradation from the synthesis of Axin ( $v_{14}$ ). The function  $f(W)$  is a decreasing function of Wnt. It is easy to see that  $\beta$ -catenin down-regulates itself through the production of APC. This mechanism is necessary to create an increase in APC along the major crypt axis as cells mature and move toward the top of the crypt. Both APC ( $X_7$ ) and Axin ( $X_{12}$ ) appear to mediate the influence of the Wnt-dependent degradation of  $\beta$ -catenin. Furthermore, the expression for  $g$  contains the influence by all members of the destruction complex. The term  $f/g$  appears in Eq. (103) [through dependence in Eq. (106)] signifying an added modulation by the terms involving GSK-3 $\beta$ , namely the destruction complex. This is all a result of increased Axin concentration within the cell.

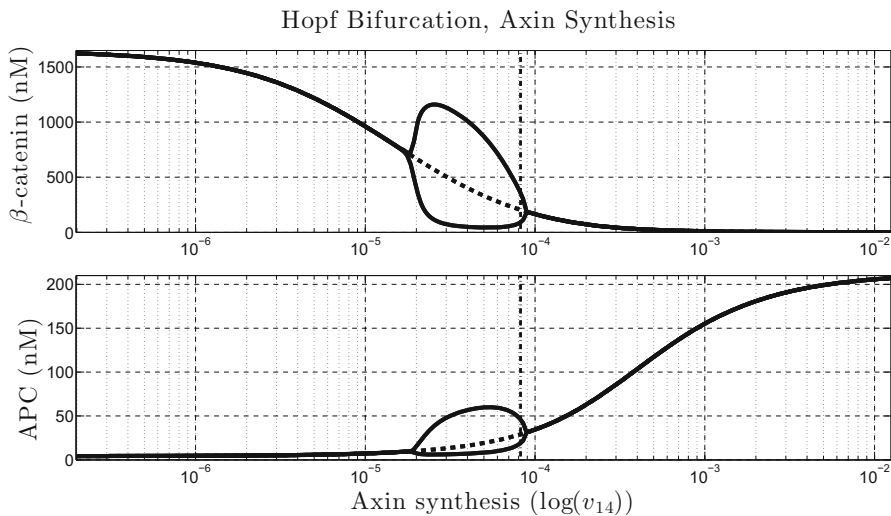
Figure 4 shows a comparison of the approximation given by Eqs. (102)–(104) to the full model subject to a transient Wnt signal,  $W(t) = \exp(-t/2500)$  for  $t > 0$ . Here, the pulse of Wnt is exaggerated so that the oscillatory behavior of the system can be demonstrated. As before,  $\beta$ -catenin, APC, and Axin reach a quasi-steady-state value before the Wnt signal is turned on. During Wnt stimulation, the approximation experiences similar oscillatory behavior as the full model; however, it is slightly shifted due to the rapid equilibrium approximations to the formation of the active destruction complex.



**Fig. 5** Phase portraits and trajectories of the reduced system [see Eqs. (102)–(104)] for several values of Wnt: (left)  $W = 1$  and (right)  $W = 0.2$

This model is interesting mathematically as well as biologically. Noting from the nondimensional simulation presented in Fig. 2, we can conclude that the system exhibits oscillatory behavior when  $W = 1$  and normal asymptotically stable behavior when  $W = 0$ . Hence, a bifurcation exists and is controlled by the Wnt signal. Using the long-term approximation given in Eqs. (102)–(104), we can plot the null surfaces of the system in  $(X_7, X_{11}, X_{12})$  space, i.e., the zero surface corresponding to the steady state solutions of Eqs. (102)–(104). Figure 5 shows the null surfaces of the system with a trajectory starting from the standard initial conditions. Each plot is for a different value of Wnt—the left figure shows a value of  $W = 1$  and the right shows a Wnt signal of  $W = 0.2$ . We can see that a periodic orbit forms around the equilibrium point for the higher value of Wnt. Furthermore, trajectories that begin near the equilibrium point will move toward the periodic orbit, indicating that instability occurs locally near the fixed point. However, for some threshold value, the orbit collapses to the equilibrium point, where damped oscillations are observed. We find that for  $W = 0.2$ , the signal is too weak to maintain the periodic orbit, resulting in a stable equilibrium point with damped oscillations. When  $W \approx 0.277$ , the protein levels begin to oscillate between two values. As  $W \rightarrow 1$ , the amplitude of these oscillations grow; however, the growth of the equilibrium value is limited. Hence, we can conclude that the Wnt signal controls the biphasic state of the cell. This describes a super-critical Hopf bifurcation and  $W$  is a control parameter. Applying this model to the colonic crypt implies that cells residing in the base experience a proliferating phase followed by a slightly shorter quiescent phase. As they migrate up the crypt and out of the Wnt signal, they experience a solely non-proliferative phase as their levels of APC begin to rise and the oscillation of protein concentrations die out.

In light of this new evidence concerning the concentration of Axin within mammalian cells, we became interested in the system's response to increased levels of Axin. The parameter,  $v_{14}$ , directly controls Axin synthesis, which determines the amount of functional Axin within the cell. Figure 6 depicts the steady state values of both  $\beta$ -catenin and APC based on various values of  $v_{14}$  on a semi-log plot. The solid line depicts the extrema of the oscillations of  $\beta$ -catenin and APC at steady state. The dashed line is the equilibrium value of the steady state solutions. If the solid line lies on the dashed line, the equilibrium point is stable. The Hopf bifurcation is clearly seen for values of  $v_{14}$  that range from  $10^{-5}$  to  $10^{-4}$ . We conjecture that any noticeable



**Fig. 6** Bifurcation diagram showing the asymptotic values of  $\beta$ -catenin (*top*) and APC (*bottom*) as  $v_{14}$  varies with  $W = 1$ . *Solid lines* indicate the extreme values of the oscillations, the *dashed line* is the equilibrium solution of the steady-state equations, and the *vertical dashed-dotted line* is  $v_{14} = 8.22 \times 10^{-5}$ , the normal level of Axin synthesis as stated by Lee et al. for the *Xenopus* egg extract

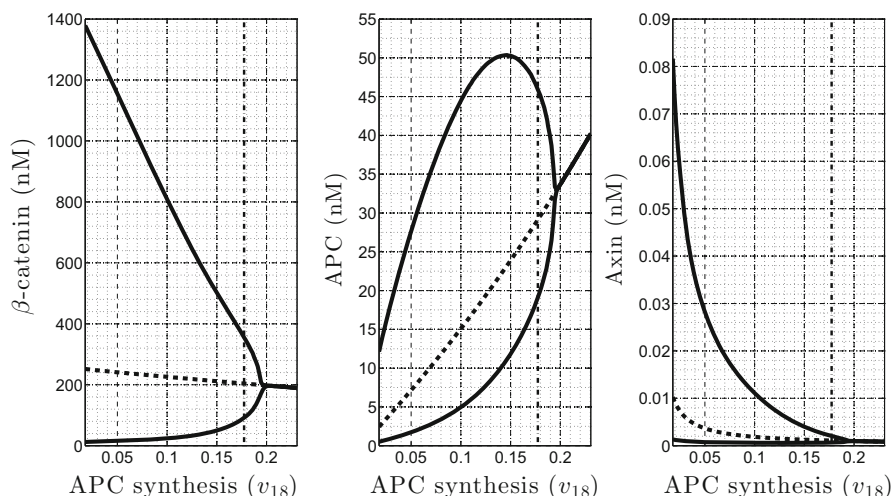
increase in Axin synthesis will drown out any oscillations in this model and ultimately result in a substantial decrease in the steady state amount of  $\beta$ -catenin, which reduces the proliferative capabilities of the cell. Further, we see that as the amount of Axin increases, the concentration of APC attains its maximum value within the cell, which is a result of there being less  $\beta$ -catenin.

We are also interested in the period and amplitude of these oscillations, and how they may depend on system parameters as perturbations in APC synthesis ( $v_{18}$ ) may lead to increased  $\beta$ -catenin levels and ultimately abnormal cell behavior. Figure 7 shows the bifurcation of steady-state behavior in the stimulated state ( $W = 1$ ) for varying APC synthesis, which determines the amount of functional APC inside the cell. Here, we can see that for higher values of APC, the system is stable, no oscillations are observed, and the result is a high level of APC with a limited accumulation of  $\beta$ -catenin. For the normal value of APC,  $v_{18} \approx 0.1774$ , there is modest oscillations in  $\beta$ -catenin; however, for a 50% decrease in APC synthesis, we see that  $\beta$ -catenin reaches a maximum near 1000 nM. Furthermore, the normal period of these oscillations is approximately 9 days. In a loss of APC functionality, the period is nearly doubled and the period increases further in a 90% decrease in APC synthesis. This indicates that in the presence of APC mutations, the crypt cell population is more proliferative and the crypt cells are in a proliferative state for a longer period of time.

### 3 Regulation of APC

In this section, we consider a special case of the previous model that isolates the effects of the APC regulatory mechanism. Here, we assume that Axin is small and present a similar reduced model of two equations at leading-order using an asymptotic multiple

## Hopf Bifurcation, APC Synthesis



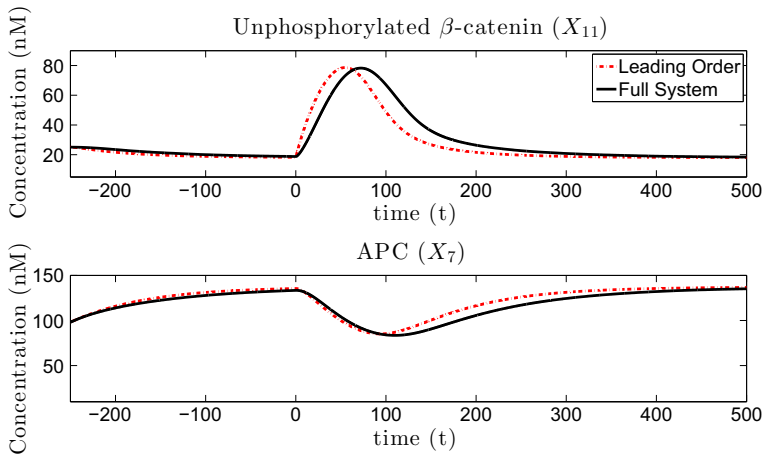
**Fig. 7** Bifurcation diagram showing the asymptotic values of APC (left),  $\beta$ -catenin (middle), and Axin (right) as  $v_{18}$  varies with  $W = 1$ . Solid lines indicate the extreme values of the oscillations, the dashed-dotted line is the equilibrium solution of the steady-state Eqs. (102)–(104), and the dashed line is  $v_{18} = 0.1774$ , the normal level of APC synthesis

timescale analysis. This model is a reduction of the previous model using the same assumptions of the Lee et al. (2003). This gives us a framework for discussion about the inverse gradients of APC and  $\beta$ -catenin that are observed in the colonic crypt (Boman and Fields 2013). We discuss an overview of the analysis and a comparison of the two models below.

We consider the original 15 equations derived in the Lee–Heinrich model with the mediated synthesis of APC by  $\beta$ -catenin and  $\beta$ -catenin/TCF, and the first-order degradation of APC [reactions 18 and 19 in Fig. 1, Eq. (35)]. Assuming Axin is small, reactions 20 and 21 are nonexistent and we assume the degradation of Axin is no longer influenced by the concentration of APC. Hence, reaction 15 is a first-order reaction given by  $r_{15} = k_{15}X_{12}$ . We consider all conservation equations and rapid equilibria approximations above. Under these assumptions, we perform an identical nondimensional analysis followed by a multiple timescale analysis as in the previous section. The short timescale result yields Eq. (81). On the intermediate timescale, we are left with a four-dimensional system of equations at leading-order. This system is composed of Eqs. (90)–(92) and, in contrast to the previous system, the evolution of Axin, given by

$$\frac{dX_0}{d\tau} = \frac{1 + (\widehat{k'_3}V_0 + \widehat{k'_{-6}})D_{i0} - (\widehat{k'_{15}} + \widehat{k'_6}P_0)X_0}{1 + \widehat{K'_7}P_0}. \quad (110)$$

The long timescale characterizes the major dynamics of the system. Here, we find that Axin is constant to leading order,  $X_0(T) = 1/\widehat{k'_{15}}$  and the only variables changing are



**Fig. 8** A comparison of the full system of nondimensional equations and the leading order solution with a transient Wnt signal given by  $W = e^{-t/20}$  and  $v_{18} = 2.26 \times 10^{-3}$ ,  $k_{19} = 4.10 \times 10^{-6}$ ,  $K_t = 8.9336$ ,  $K_b = 42$

$\beta$ -catenin ( $B_{a0}$ ) and APC ( $P_0$ ). We present the equations in the original parameters below:

$$\frac{dX_7}{dt} = \frac{v_{18}}{1 + \frac{TCF^0 X_{11}}{K_7(K_{16} + X_{11})} + \frac{X_{11}}{K_b}} - k_{19}X_7 \quad (111)$$

$$\frac{dX_{11}}{dt} = \frac{v_{12} - \left[ k_{13} + f(W)X_7 + \frac{1}{K_{17}} \frac{dX_7}{dt} \right] X_{11}}{1 + \frac{X_7}{K_{17}}}, \quad (112)$$

where

$$f(W) = \frac{k_1 k_4 k_6 k_9 v_{14} GSK^0}{k_5 k_{15} K_7 K_8} \left( \frac{W + k_2/k_1}{k_1 (Dsh^0 k_3 + k_{-6})W + k_2 k_{-6}} \right). \quad (113)$$

We can see that there are three types of  $\beta$ -catenin degradation: a free form destruction ( $k_{13}$ ), a Wnt and APC mediated degradation by the destruction complex ( $f(W)X_7$ ), and an additional decrease by the change in APC. The term  $f(W)X_7$  and the additional degradation due to the change in APC characterize the main difference from the analysis of Mirams et al. (2010). For example, if APC is conserved, then  $X_7 = APC^0$ , and the leading-order approximation for  $X_{11}$  of Mirams et al. comes through in the analysis. The function  $f(W)$  is a decreasing function of Wnt. In the reduced model, it is easy to see that  $\beta$ -catenin down-regulates itself through the production of APC. This mechanism is necessary to create an increase in APC along the major crypt axis as cells mature and move toward the top of the crypt.

Figure 8 shows a comparison between the leading order solutions given by Eqs. (111) and (112) and the solutions to the full system subject to a transient Wnt signal,  $W(t) = \exp(-t/20)$ . We note that the initial amount of  $\beta$ -catenin peaks in the presence of a diminished initial amount of APC but is quickly degraded as the amount of APC

is raised. This leads to a quasi-steady state in both the full system and leading order solution before the Wnt signal is activated. During the Wnt stimulus, we can see that the leading order solution acts faster and achieves the maximum more quickly than the original system. This is because reactions occurring on the fast timescales are approximated by rapid equilibria. However, the long term dynamics of the leading order solution capture the full system solution well.

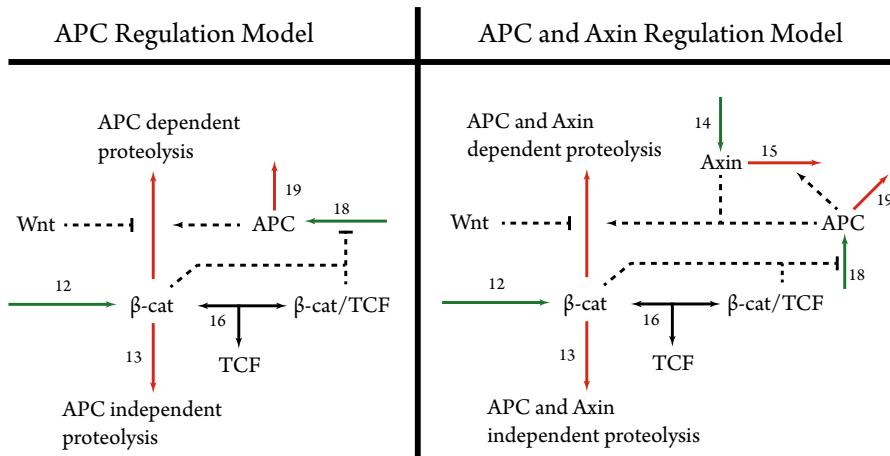
We note here that the system given by Eqs. (111) and (112) has a unique positive equilibrium point whose linear stability can be tested numerically (not shown). We were able to confirm the point is asymptotically stable for any choice of parameters  $v_{18}$ ,  $k_{19}$ ,  $K_t$ , and  $K_b$  in both the Wnt stimulated and unstimulated state.

From this output, we can conclude that this system yields an inverse relationship between APC and  $\beta$ -catenin, which is completely controlled by the Wnt signal. In the absence of Wnt, which is the state at the top of the colonic crypt, a higher concentration of APC is present. In contrast, at the base of the crypt the Wnt signal is strong, which means Wnt target factors like  $\beta$ -catenin are more abundant in the cell. Therefore, the regulatory mechanism employed in this section will create inverse gradients of  $\beta$ -catenin and APC if subjected to a spatially varying Wnt signal.

## 4 Discussion

This study was built on and expanded the Lee–Heinrich model, which focuses on the key elements involved in Wnt signal transduction. Our novel extensions are characterized by the inclusion of an important APC regulatory loop that describes the inverse relationship between APC and proliferation promoting Wnt factors, such as  $\beta$ -catenin, that are typically observed in epithelial cells of the human colonic crypt. Our first model addresses the fact illustrated by Tan et al. that protein levels observed in the *Xenopus* egg extract do not concur with levels found in mammalian cell cultures (Tan et al. 2012). Therefore, the APC and Axin regulation model does not assume that the concentration of Axin is small compared to other quantities, which provides a more accurate framework for modeling Wnt signal transduction in the human colon. The special case of APC regulation considers the interaction between APC and  $\beta$ -catenin that uses the parameter values and assumptions of Lee’s original model (Lee et al. 2003), along with our fitted parameters for  $K_b$  and  $K_t$ . This model describes the dynamic interplay between APC and  $\beta$ -catenin that is observed in the colonic crypt and will be used in future models on the cellular level.

Indeed, the reduced leading order systems constructed in our study allows us to include the intracellular dynamics of the Wnt pathway in larger, multi-scale models of the colonic crypt, which is a topic of our next study. Furthermore, this allows us to extract the important parameters responsible for controlling the long-term dynamics of Wnt signal transduction. The APC regulation model provides the desired inverse relationship between APC and  $\beta$ -catenin using the original parameters of the Lee–Heinrich model while implementing the fitted parameters of our additional feedback loop. From this, we can gauge the influence of the APC–Wnt counter-current-like mechanism (Boman and Fields 2013) at the crypt level. Figure 9 shows the reduced schematic of reaction kinetics for the simplified APC and Axin regulation model and



**Fig. 9** (Left) a reduced schematic illustrating the reaction kinetics of the reduced, long term dynamics defined by Eqs. (111) and (112), which describes the APC regulation model. (Right) a reduced schematic that illustrates the reduced, long term dynamics defined by Eqs. (102)–(104). As before, *solid arrows* depict reactions in the indicated direction, *double-sided arrows* are binding reactions, and *broken arrows* denote mediated reactions

the APC regulation model of Sects. 2 and 3, respectively. The drawing (right) illustrates the core reactions that induce the system into a periodic orbit.  $\beta$ -catenin and  $\beta$ -catenin/TCF act as an inhibitor to APC production. APC inhibits Axin by positively reinforcing its degradation. Finally, APC and Axin enhance  $\beta$ -catenin destruction by forming the destruction complex. Hence, the system operates as an oscillatory network driven by three inhibitory responses. Not included in the figure are the binding reactions associated with Axin and APC, namely binding kinetics involving dissociation constants  $K_{20}$  and  $K_{21}$ . We find several bifurcation parameters including the Wnt signal ( $W$ ), APC production ( $v_{18}$ ), and Axin synthesis ( $v_{14}$ ), which influence the period and amplitude of these oscillations as discussed in Sect. 2. The reduced model can be used to describe the dynamics of the Wnt pathway without assuming Axin levels are low. Overall, the timescale analysis of this model results in a three equation system that acts as a basis for inclusion in multi-scale models and in modeling Wnt signal transduction in the human colon.

The behavior of the intracellular, biochemical model of APC and Axin regulation provides interesting interpretations at the cellular level. As noted by Lewis et al., the dual role of Axin may lead to oscillations in the levels of  $\beta$ -catenin (Lloyd-Lewis et al. 2013). Our model simulates Axin's behavior as it responds to the Wnt signal by incorporating the interaction between APC and Axin. The mediated inhibition of Axin synthesis by APC essentially characterizes the feedback proposed by Cho et al. (2006), although it acts through APC first and inhibits degradation of Axin instead of positively reinforcing its synthesis. In this sense, Axin's dual role is a result of it being recruited to the destruction complex when APC is low. Assuming a cell's phenotypical properties are based on intracellular  $\beta$ -catenin levels, the cell experiences a biphasic state, where the cell is either proliferating or quiescent. In the colonic crypt, this means cells at the base (those exposed to a higher level of Wnt) are constantly alternating

between the proliferating and non-proliferating state. Crypt renewal, then, may be related to waves of cells originating at the base and migrating up lending to the idea that crypt fission may be initiated when periodic cell proliferation cycles attain a maximum. Indeed, accelerated crypt fission is thought to be a factor in the expansion of colorectal adenomas (Boman and Fields 2013). The oscillatory behavior of this model agrees with biological observations in Lloyd-Lewis et al. (2013), Orford et al. (1997), Cliffe et al. (2003) and explains the initiation of crypt fission as conjectured by Boman and Fields (2013). That is, when a peak in intracellular  $\beta$ -catenin levels occurs in the cell at the base of the crypt, an increase of cell proliferation and division occurs, which effectively initiates the budding phenomenon before a wave of new cells populate the new crypt. Under an APC mutation, i.e., a decrease in APC synthesis (see Fig. 7), this process may be accelerated.

We note here that the APC–Axin regulation approximation holds up when subjected to large initial quantities of Axin and when a larger synthesis rate,  $v_{14}$ , are implemented in the system; however, oscillations do not occur in the latter case. This is expected as APC and  $\beta$ -catenin rates must also be scaled appropriately to maintain the oscillation studied in Sect. 2. We perform a sensitivity analysis on the APC regulation model (not shown) to test the system's response to single perturbations in model parameters. Similar to that of Kruger and Heinrich (2004), results show that the system remains stable when exposed to a 1% adjustment in system parameters. The system is most sensitive to perturbations in Axin synthesis even with the APC feedback loop.

Future work is needed by both mathematicians and biologists to validate this model and enhance its relevance to the true dynamics that occur within the cell. The APC–Axin regulation model presented here is open to interpretation by biologists and model parameters such as the Axin synthesis rate should be calibrated to mammalian cell lines through biological experimentation. In our study, the parameter values that we used were based on the original study of Lee et al. (2003) on *Xenopus* eggs with the exception that we used the concentration of Axin based on the study of Tan et al. (2012), which examined five mammalian cell lines. It also revealed that in colon cancer cells with mutant APC, the APC concentrations are higher than Axin concentrations. Additional biologic studies have been done on Axin and APC because both are large scaffold proteins required for the assembly, stability, and activity of the Wnt signaling destruction complex. Much of this work has involved investigation of the ratio of APC to Axin and how it influences assembly of the destruction complex as well as efficiency of beta-catenin degradation (Kunttas-Tatli et al. 2014; Pronobis et al. 2015). While other studies have clarified which components exist in the destruction complex (MacDonald et al. 2009), additional biologic work needs to be done on understanding the concentrations of these proteins and how this influences Wnt signaling. Once parameter values are agreed upon, the model can be used to simulate and predict various scenarios that may occur between protein concentrations in the cell.

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